

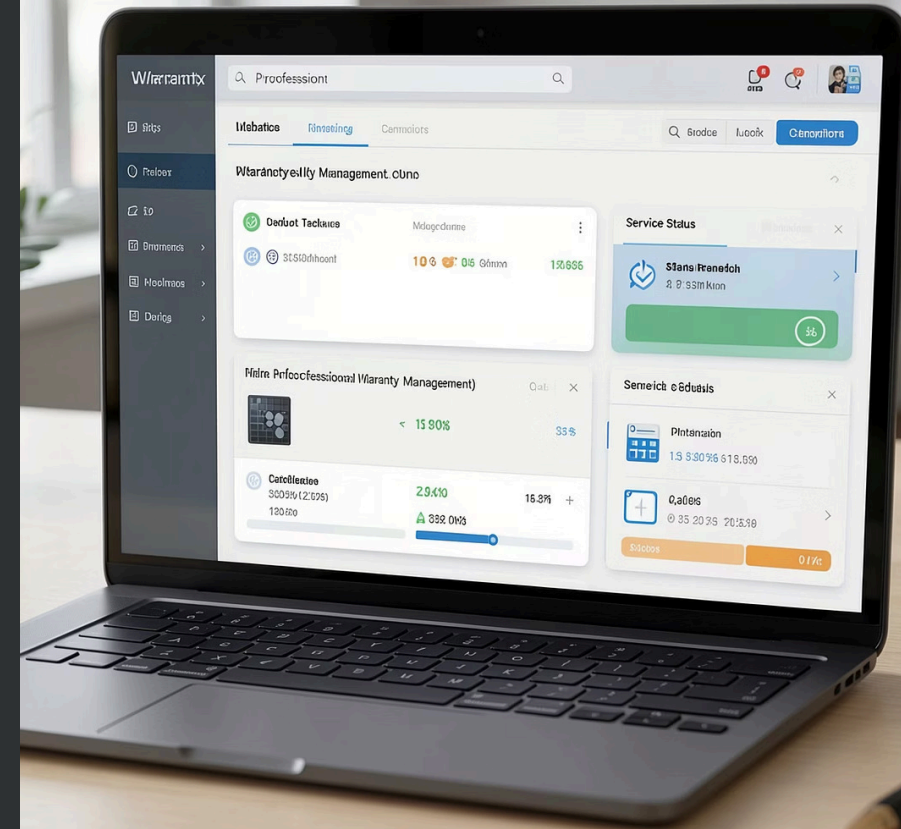
Smart Warranty & Product Service Tracker

IIT Mandi Capstone Project

https://github.com/Tulika1910/Smart_Warranty_and_Product_Service_Tracker/tree/main/Capstone_Project

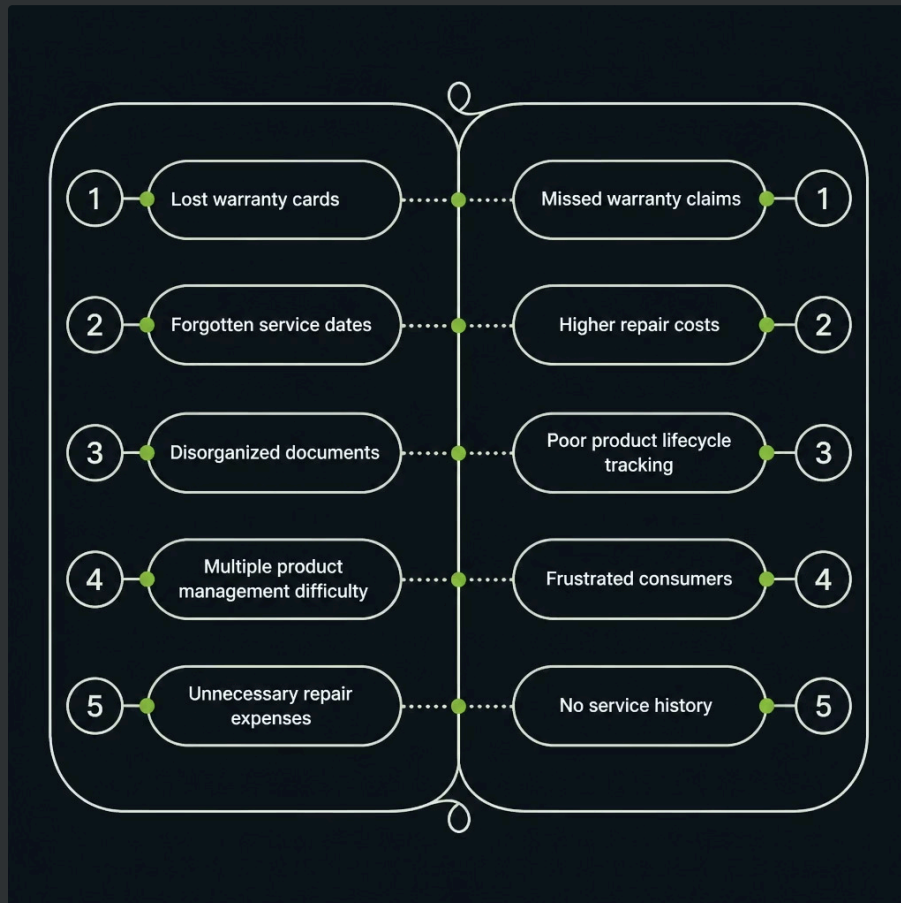
Team Members

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The Warranty Management Problem

Millions of consumers lose track of warranties every year — leading to missed service claims and unnecessary repair costs. Without a centralized system, warranty documents become scattered, forgotten, or lost entirely.



Why This Matters

Warranty cards are physical and easily lost

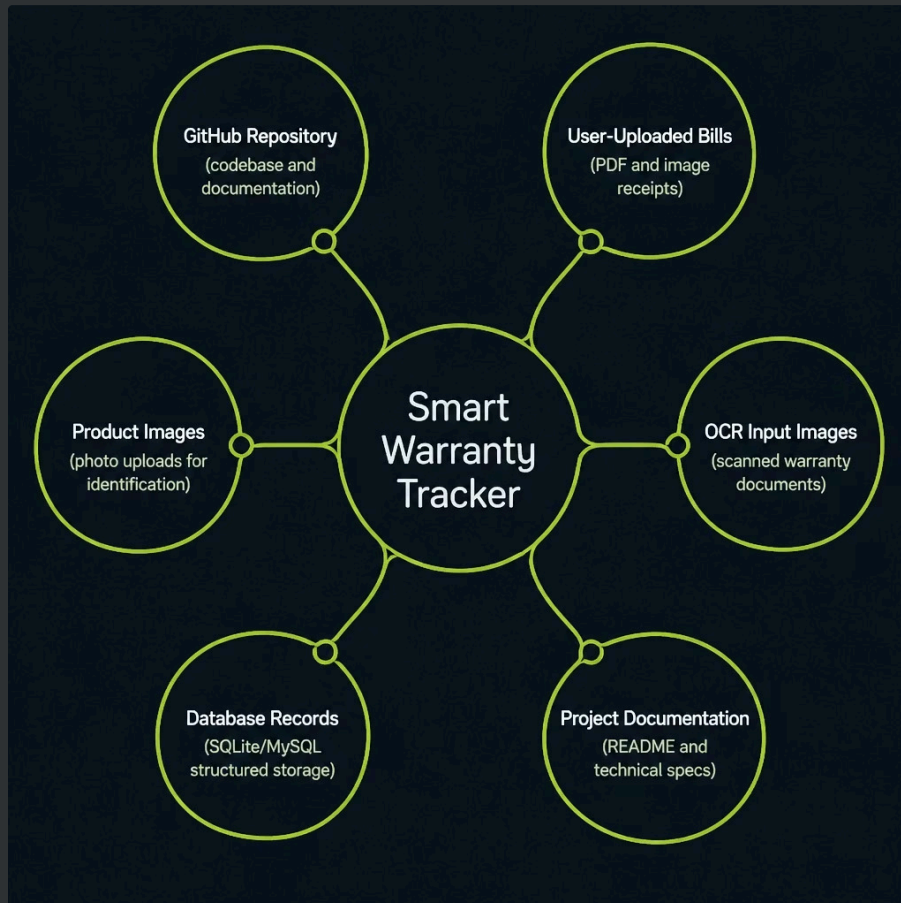
Service dates go untracked without reminders

No centralized view of all owned products

Missed claims result in avoidable expenses

Project Inputs & Reference Sources

The system draws from multiple structured and user-provided data sources to enable intelligent warranty extraction and tracking.



Data Sources



GitHub Repository

Full source code, README, and project documentation



User-Uploaded Bills

Purchase receipts and invoices submitted by users



Product Images & OCR

Scanned documents processed through OCR pipeline



Database Records

Structured SQLite/MySQL storage for all warranty data

End-to-End System Workflow

From login to notification — every step is automated, secure, and user-friendly.

01	02	03
Login	Add Product	Upload Warranty Document
04	05	06
Store Product Information	Database	Dashboard
07	08	09
Warranty Tracking	Service Reminder	Chatbot Assistance

The pipeline processes user-uploaded bills through OCR, extracts key warranty fields automatically, persists data to the database, and triggers timely service reminders — all without manual data entry.

AI & Software Innovation



AI Chatbot

Natural language processing for warranty queries and product assistance



Streamlit Frontend

Python web framework for building the interactive user interface and dashboard



Database Management

SQLite/MySQL storing warranty, product, and user data



User Authentication

Secure login with session management and access control



Warranty Tracking

Real-time monitoring of warranty expiration dates



Service Reminder System

Automated alerts for upcoming service dates

Tech Stack & Dependencies

Core Libraries

```
import os
import uuid
from datetime import date

import google.generativeai as genai
import pandas as pd
import plotly.express as px
import pymupdf
import streamlit as st
from PIL import Image
from sqlalchemy import text
from streamlit_mic_recorder import speech_to_text
```

Core libraries and APIs powering the application

Key Technologies Used



Gemini API

Google's generative AI for chatbot functionality



Firebase Authentication

Secure user login and session management



Streamlit

Interactive web framework for building the user interface



Pandas

Data manipulation and analysis capabilities



Plotly Express

Interactive visualizations for dashboard insights



SQLAlchemy

Database Object-Relational Mapper (ORM) for data persistence



PIL/Pillow

Image processing library for handling product photos and scans



PyMuPDF

Efficient PDF handling for user-uploaded bills and documents

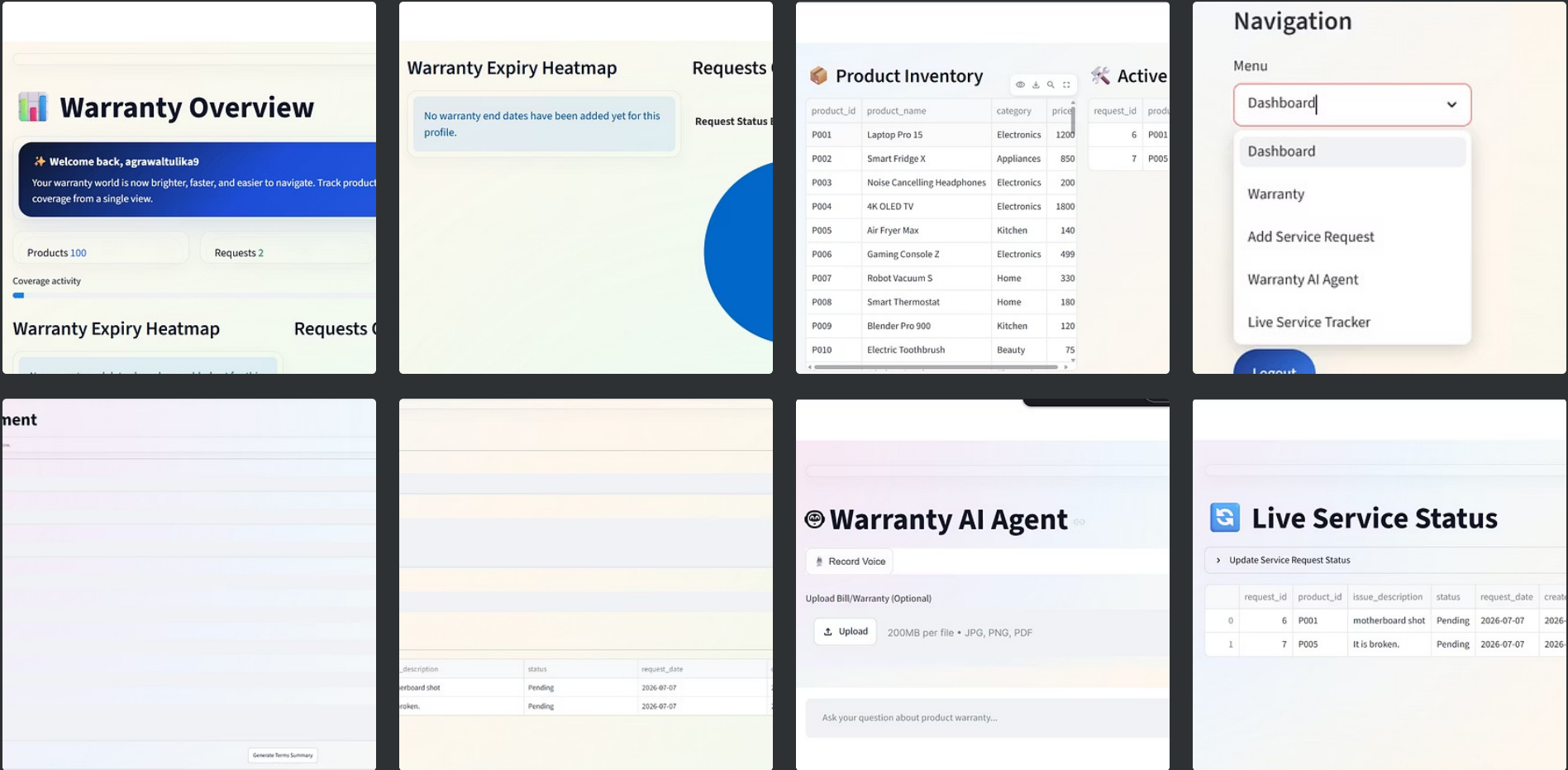


Streamlit Mic Recorder

Enables voice input for the AI chatbot

Prototype Screenshots

Key screens from the live application demonstrating the complete user journey from authentication to warranty management.



Results & Sample Outputs

Dashboard & Tracking

Product List

All registered products displayed with warranty status badges

Warranty Details

Click-through view showing purchase date, expiry, and vendor info

Service Reminders

Automated alerts triggered before warranty expiration dates

Limitations & Responsible Use

Current Limitations

OCR Quality Dependency —
Extraction accuracy depends on bill image clarity and format

Manual Verification —
Some fields may require user confirmation post-extraction

Format Constraints —
Limited support for non-standard bill layouts

Internet Dependency —
Cloud-based OCR requires active connectivity

Responsible Use Guidelines

User Privacy —
Personal purchase data must be protected and never shared

Secure Storage —
Warranty documents should be encrypted at rest

Access Control —
Strong authentication prevents unauthorized document access

Ethical Use —
Uploaded documents must not be misused or redistributed

Future Roadmap

- 1 Phase 1 — Core Enhancements**
Improved OCR accuracy, barcode scanner integration, QR code support for product registration
- 2 Phase 2 — Mobile & Cloud**
Cross-platform mobile app (iOS/Android) with cloud synchronization for multi-device access
- 3 Phase 3 — AI Features**
AI chat assistant for warranty queries, predictive maintenance alerts, and smart recommendations
- 4 Phase 4 — Global Reach**
Multi-language support, regional vendor integrations, and enterprise deployment options



Conclusion



Digitized

Replaces physical warranty cards with a secure digital system



Organized

Centralized dashboard for all products and warranty documents



Proactive

Automated reminders prevent missed service and warranty claims



Secure

Authentication and encrypted storage protect user data

This project demonstrates practical software engineering — combining OCR, web development, and database design to solve a real consumer problem.

Thank You

Himshikar Capstone Project · Smart Warranty & Product Service Tracker

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